# AudioConditioner

Models & Training

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CSC_52002_EP - Multimodal generative AI

## INTRODUCTION & THE DESCRIPTOR MODEL

The **AudioConditioner** is a high-level multimodal model that bridges the gap between visual/textual semantics and music generation utilizing **BLIP** for image captioning and the **CLAP** text encoder for semantic embeddings.

The core innovation is our **Descriptor Model**, which is the *only trainable component in the entire architecture*. All other models (BLIP, CLAP, Stable Audio) are kept completely frozen. The Descriptor acts as a multi-task learning bridge, mapping the CLAP text embedding through a shared neural backbone into 12 discrete and continuous musical attributes.

The generated attributes are structurally combined within a `MusicDescriptor` object. This object then programmatically generates the final descriptive text prompt fed into **Stable Audio** (which has a 47-second generation limit). Finally, to ensure strict semantic alignment, we generate $N$ candidates and re-rank them using **Cosine Similarity** against the CLAP text embedding.

## THE MUSICDESCRIPTOR SCHEMA

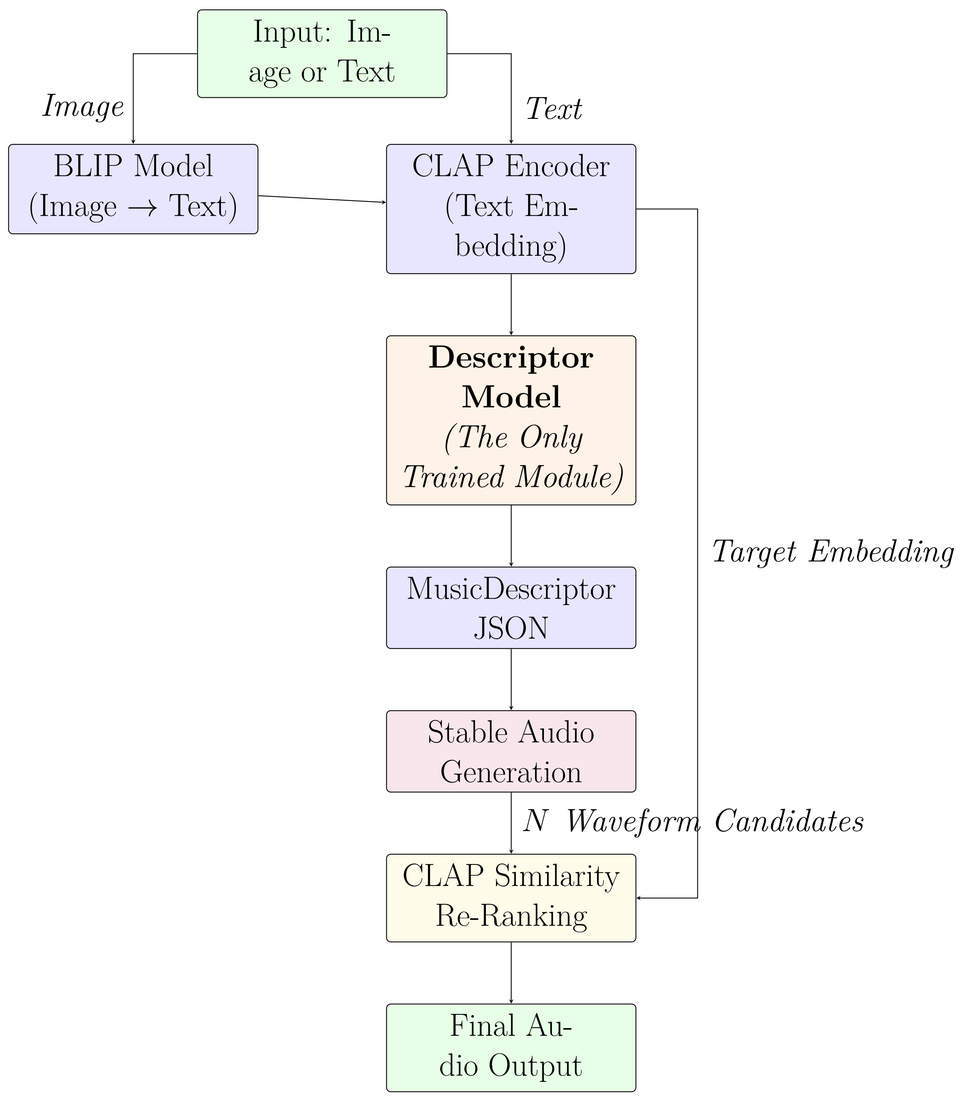
### Classification Attributes

- **Mood** *(→ Multi-Label List)*: Emotional states (e.g., dark, epic, melancholic).
- **Key/Mode** *(→ Single-Label)*: The musical scale used (e.g., major, minor).
- **Instrumentation** *(→ Multi-Label List)*: Active instruments (e.g., strings, synth).
- **Rhythm Style** *(→ Single-Label)*: Overall rhythmic feel (e.g., sparse, pulsing).
- **Structure** *(→ Single-Label)*: Internal song progression (e.g., slow-build).
- **Production Style** *(→ Multi-Label List)*: Mixing/Genre aesthetic (e.g., cinematic).
- **Dynamics Profile** *(→ Single-Label)*: Volume progression (e.g., gradual build).

### Regression Attributes

- **Energy** *($[0.0, 1.0]$)*: Overall intensity level (0.0 = very calm, 1.0 = very energetic).
- **Valence** *($[0.0, 1.0]$)*: Emotional polarity (0.0 = very negative/sad, 1.0 = very positive/happy).
- **Tempo** *($[50, 180]$ BPM)*: Beats per minute, strictly normalized to $[0.0, 1.0]$.
- **Harmonic Tension** *($[0.0, 1.0]$)*: Chord dissonance (0.0 = very consonant, 1.0 = very dissonant).
- **Texture Density** *($[0.0, 1.0]$)*: Number of simultaneous elements (0.0 = very minimal, 1.0 = very layered).

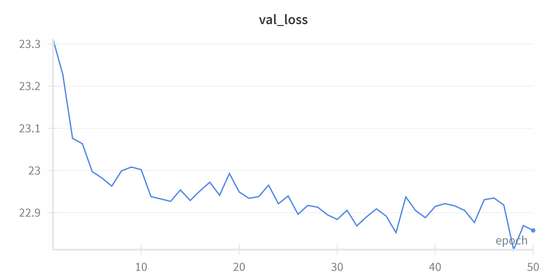
## FRAMEWORK ARCHITECTURE

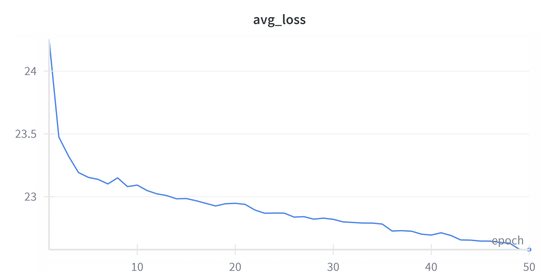



## PHASE 1: GENERATIVE LLM DISTILLATION

Training the **Descriptor Model** required a large parallel corpus mapping natural language to the strict `MusicDescriptor` format. We built this automatically through **Prompt Mixing** and **Teacher LLM Distillation**:

- **Data Generation:** Instead of using an LLM to write scenes, we procedurally generated millions of short scene combinations by randomly mixing predefined *characters*, *characteristics*, *settings*, and *emotional events* (e.g., *"A lonely astronaut under a burning sunset discovers an unexpected opportunity."*).
- **Distillation:** We then prompted an LLM to act as an expert composer, taking these random scenes and outputting the deterministic JSON `MusicDescriptor` (Mood, Tempo, etc.) fitting each scene.
- **Optimization:** The model was trained using an `AdaptedMusicDescriptorLoss` function (Cross-Entropy for categorical attributes and MSE for continuous target regressions).



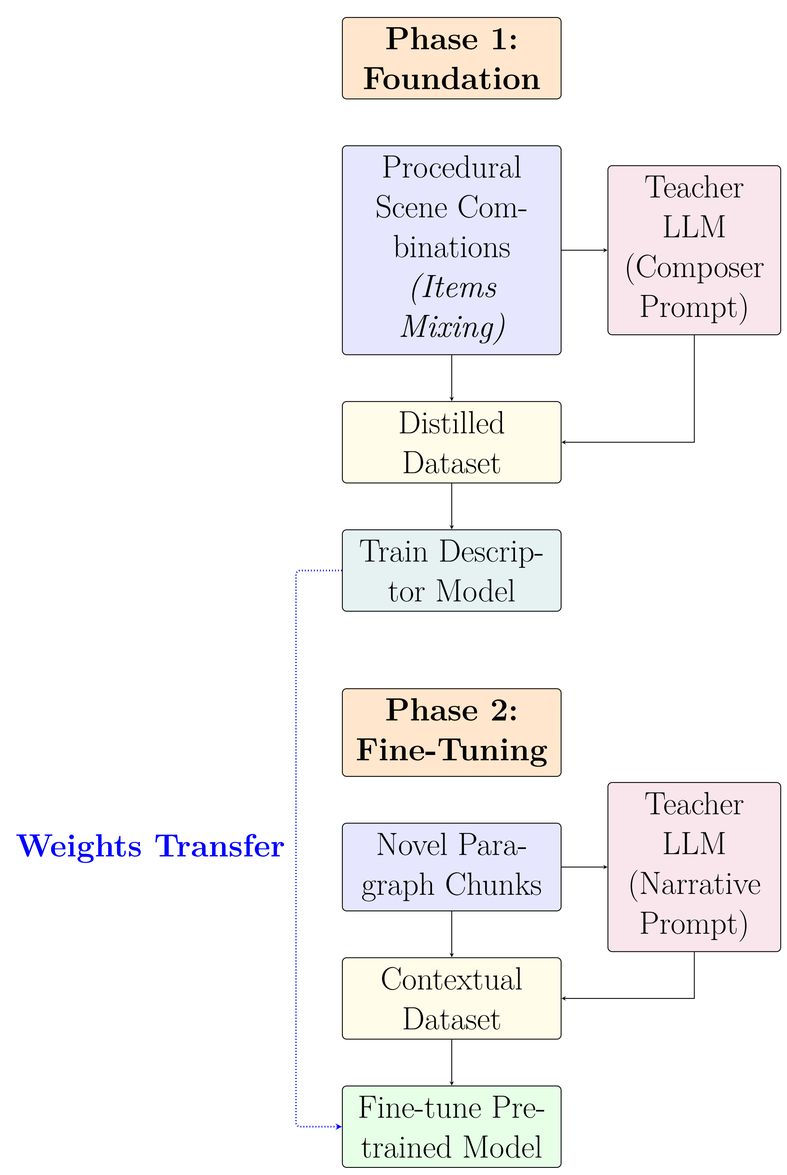




## PHASE 2: CONTEXTUAL NOVEL FINE-TUNING

To support narrative audiobook generation (**ReadMusic**), we fine-tuned the model to handle longer contextual evolution:

1. **Novel Chunking:** Public domain novels data were split into consecutive paragraphs (up to 80 words max).
2. **Target Generation:** The Teacher LLM assigned progressive `MusicDescriptors` respecting the storyline's emotional flow.
3. **Fine-Tuning Outcomes:** The Descriptor internalizes literary context, preventing abrupt changes and enabling smooth, continuous emotional transitions for long-form reading.

## TRAINING PIPELINE OVERVIEW

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